

ThermoTwin architecture

Hybrid physics + ML twin: the medication record changes the physiology model, not just the score

DATA STREAMS

Static EHR (synthetic)

BP meds by class, eGFR history, NSAIDs, age, BMI

Wearable stream

1-min heart rate + activity (JOS-3 simulated shifts)

Weather

Open-Meteo archive: Delhi heatwave, May 2024

Real heat trials

PROSPIE (22 people, 99 trials): validation only

TWIN ENGINE (PYTHON)

1 ECTemp Kalman filter

Core temperature from heart rate

2 Medication correction

β -blocker: un-blunt HR, widen noise

3 Personal bias learning

Thermometer readings from past shifts

4 60-min forecaster

Gagge physics + gradient boosting

SERVICE

FastAPI

/patients
/timeline
/whatif
/review

Review rules

Triple whammy, eGFR decline, β -blocker masking, heat strain

USERS

Dashboard

React: live twin, what-if, review

PHC doctor

Pre-summer medication review

Deploy: Docker Compose (non-root API with health check · nginx serves dashboard and proxies /api) · Quality: pytest suite, participant-held-out real-data validation, multi-seed evaluation